

THE PENNY MAGAZINE



Issue 13

June 16, 1832

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Society for the Diffusion of Useful Knowledge, issue 13, June
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Title: The penny magazine of the Society for the Diffusion of Useful
Knowledge, issue 13, June 16, 1832

Editor: Charles Knight

Release date: September 4, 2025 [eBook #76818]

Language: English

Original publication: London: Society for the Diffusion of Useful
Knowledge, 1832

Other information and formats: www.gutenberg.org/ebooks/76818

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MAGAZINE OF THE SOCIETY FOR THE DIFFUSION OF USEFUL
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OF THE

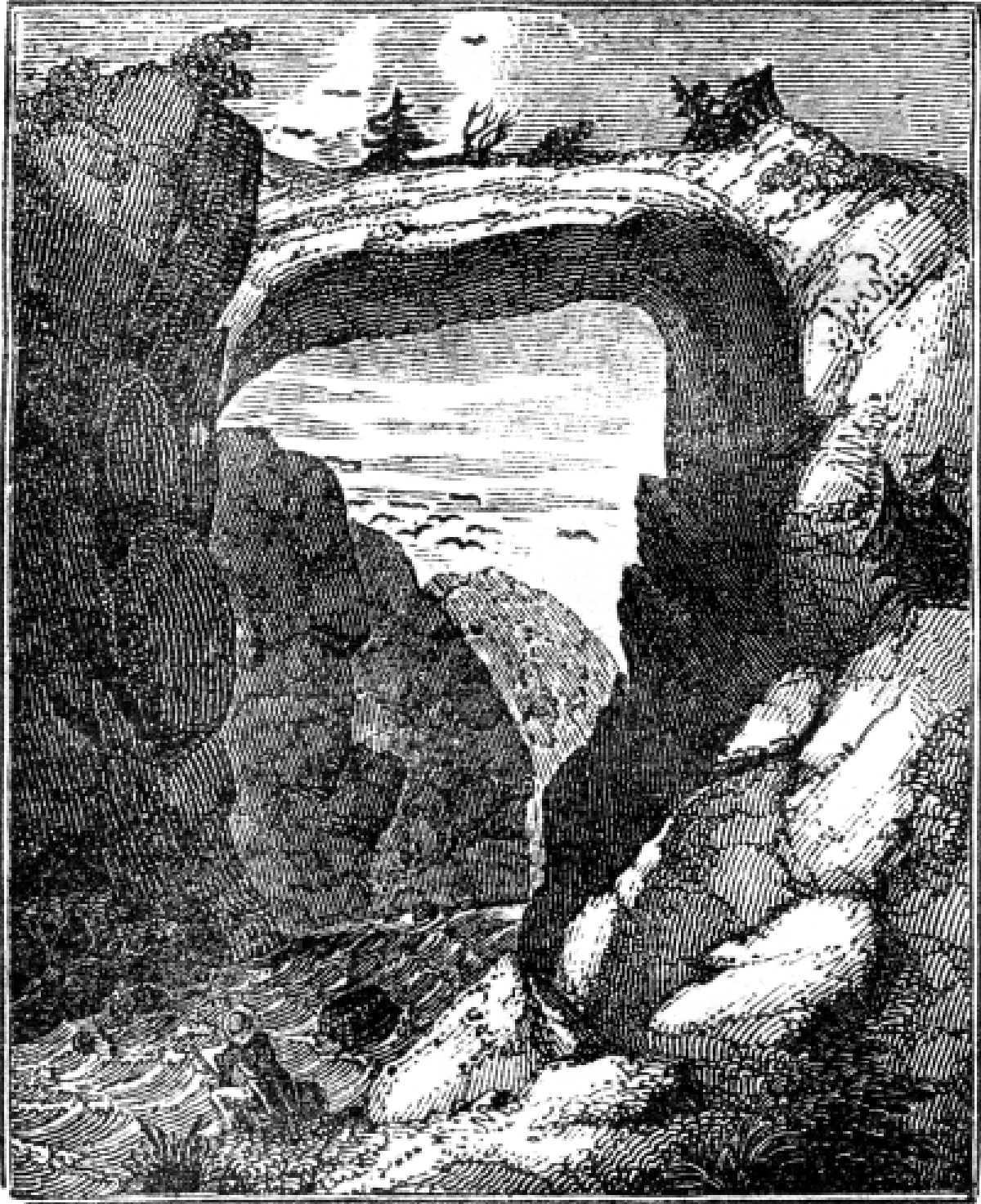
Society for the Diffusion of Useful Knowledge.

13.]

PUBLISHED EVERY SATURDAY.

[JUNE 16, 1832

THE NATURAL BRIDGE OF VIRGINIA.



Virginia, the largest state in the American Union, is intersected by a chain of mountains called the Blue Ridge, which, running in their general direction parallel to the Atlantic coast, divide the state into two parts not

differing very considerably in extent. The portion immediately to the west of the Blue Ridge is an extensive and fertile valley of limestone formation. It is principally watered by one stream, the Shenandoah, which unites with another, the Potomac, at a place called Harper's Ferry. At their point of junction, on the west side of the Blue Ridge, the spectator, as he takes his stand on the high ground above the small town of Harper's Ferry, sees before him a wide opening in the mountain chain through which the united current finds its way. On each side the mountains rise in some parts very abruptly, and their rugged faces and the shattered appearance of the whole of this magnificent natural canal show evident traces of a violent disruption.

This passage at Harper's Ferry has been often described by different travellers, but never, as far as we have seen, in a way calculated to give an accurate conception of what it really is. Nor do we intend to attempt this description, but only to notice briefly another natural phenomenon of the Valley of Shenandoah, which, though less talked of and visited than Harper's Ferry, is for beauty and grandeur perhaps unrivalled. We allude to the Natural Bridge, or Rock-Bridge, as it is familiarly called by the people who live near it, which is situated a few miles on the west side of the Blue Ridge, on a small stream in the upper part of the great valley, and in the county of Rock-Bridge.

From a small and uncomfortable tavern in the neighbourhood, kept by a Mr. Galbraith, (we wish this could meet his eye and make him mend his fare,) we pass for about two miles over uneven ground, and after ascending a small hill, we find a piece of rough stony road with a few stunted firs and scrub oaks on the right hand and on the left. A traveller might proceed without making any other observation, as the common road runs right over the bridge, and it is said that some people have actually passed over without being aware of it. But though this is certainly a possible occurrence if a person should be in a closed carriage, it can hardly have happened to a man on foot or on horseback, who is accustomed to keep his eyes open when he is travelling. On the right and left he will perceive that the slope of the hill is interrupted by a deep and sudden descent; and on going nearer to the right side of the road, he finds himself on the edge of a tremendous precipice. At the bottom a small stream is seen making its way amidst broken rocks. Going to the opposite side of the road and looking down there, he will observe the little river continuing its course in a deep channel down a narrow valley. The traveller is now on the Natural Bridge; he is

standing on a stupendous natural arch of limestone; and though he may form some conjecture of his situation by looking down from the edge of the precipice, he can have no adequate conception without viewing it from below. The arch is best seen from the bed of the rivulet, and from a point just under it. On looking up you behold a noble arch of one solid mass of stone hanging over your head, somewhat curved in its highest part, and almost like the work of man. The same native rock forms, on each side, the supports of this enormous arch, which is said to be about 80 feet wide near the top; at the level of the water the width is only about forty. The whole height from the outer top of the arch to the water is about 210 feet, as ascertained by measurement with a string and a stone at the end. This is greater than the height of the London monument. The vertical thickness of the arch is probably about 30 feet. Like many other great works both of nature and art it is not the first sight that produces the deepest impression. On a second visit we found that we had learned to form more accurate conceptions of this wonderful bridge, beneath which a man might sit and gaze for hours with still increasing astonishment at the majestic arch which nature constructed before man began his work, and which seems likely to outlive the most durable of his monuments. Whatever may have been the origin of this bridge, it seems pretty certain, from an inspection of it, that it has not been produced by any sudden and violent cause.

The stream that runs beneath, called Cedar Creek, though inconsiderable, adds to the general effect. When we visited the place, drops of water, filtered through the limestone, were falling in quick succession from the arch, and by the time occupied in their descent, their increasing velocity, and their full bright appearance, served to give a measure of the height from which they fell, and to increase the beauty of the scene. There is another natural bridge in Virginia, in Scot county, which is said to be above 340 feet high, but is inferior to that of Cedar Creek in form and completeness.

The Prebischthor, in the Saxon Switzerland, has sometimes been compared with this Virginia Bridge, but it is a very different kind of thing.

The accompanying view, taken from the N.W. side, at the level of the water, has hardly any pretensions beyond showing the general shape of the arch and the view through it, which is very confined and altogether devoid of interest.

The chain^[1] of the Andes in South America presents most striking natural phenomena in the immense clefts, or *crevasses* as they are sometimes called, which separate two contiguous masses of mountain, and in some instances are near 5000 feet deep. If Mount Vesuvius were plunged into one of these frightful abysses, its summit would not reach to the peaks of the highest rocks on each side; while the bottom of the cleft would be only one-fourth less elevated above the level of the sea than the passes of St. Gothard and Mont Cenis in the Alps.

The valley of Icononzo is less remarkable for its dimensions than for the extraordinary form of its rocks, which seem as if they had been cut by the hand of man. Their naked and arid summits form a most picturesque contrast with the tufts of trees and herbaceous plants which cover the borders of the *crevasse*. A little torrent has made itself a way through the valley, and lies sunk in a channel, which is so difficult of approach, that the river would hardly be passable if nature herself had not formed two bridges of rock, which are justly regarded as the greatest curiosity in that country. Humboldt and Bonpland crossed these natural bridges in 1801, on their route from Santa Fé de Bogota to Popayan and Quito.

In the valley of Icononzo the *grès* or sandstone is composed of two distinct kinds of rock—one very compact and quartzose without any marks of fissure or stratification—the other a fine-grained sandstone, formed of an infinite number of thin and almost horizontal layers. We may imagine that the compact material resisted the force which rent the mountains asunder, and that it is the unbroken mass of this rock which forms the bridge by which the traveller now crosses from one side of the valley to the other. This natural arch is about 47½ feet long, 41½ wide, and about 8 feet thick at the centre. By very careful experiments made on falling bodies, with the assistance of a good chronometer, combined with the measurement obtained by a plummet, it appears that the height of the upper of the two natural bridges, above the level of the torrent, is about 313 feet.

Sixty feet below the first natural bridge there is another formed by three enormous masses of rock, which have fallen in such a way as to support one another. The centre rock forms the key of the arch.

¹. Humboldt, *Vue des Cordillères*, &c. 8vo. Paris.

STATISTICAL NOTES.

ENGLAND AND WALES (CONTINUED)

(20.) Of the state of English agriculture in early ages some notion may be formed from the fact of the prohibition for many years, and subsequently the taxation, of the exportation of corn. It was not till the reign of Charles II that the export of corn was exempted from a tax; and it is from 1689 that may be dated that fundamental change in our corn-laws which encouraged exportation by a bounty. Since that period the fluctuations in the price of corn have been remarkable. The price of wheat which in the beginning of the last century was 50s. the quarter, became reduced in the ten years between 1740 and 1750 to 24s. the quarter. The culture of corn thus received a check, and a large proportion of arable land was transferred from tillage to grazing. The effect of this conversion and of an increasing population raised the price of corn in the ten years from 1750 to 1760 to an average of 42s. 6d. per quarter, and soon changed the scale from export to import, which has continued ever since. From 1764 to 1790 the average price of wheat varied from 42s. to 50s.; our annual imports from 200,000 to 500,000 quarters of corn. But since 1792 our annual imports, under differently regulated systems of law, have been from half a million to above two million quarters of corn of all kinds; and the average prices of wheat have varied from 2l. to 6l. per quarter. In 1792 the price of wheat was 2l. 2s. 11d.; in 1800, 5l. 13s. 7d.; in 1812, 6l. 5s. 5d.; in 1822, 2l. 4s. 1d.; and in 1831, 3l. 10s. 3d. The annual consumption of wheat in the United Kingdom has been estimated at 12,000,000 quarters; and that of other grain at 36,000,000 quarters, making together 48,000,000, of which not one-twentieth part has during any year been imported, and, in general, a far less proportionate quantity. The daily consumption of wheat in the United Kingdom may be taken at 36,000, and of all other grain at 108,000 quarters, making together 144,000 quarters a day.

(21.) During the last century, upwards of five millions of acres in England and Wales have been enclosed under Acts of Parliament, the average extent of each enclosure being 1200 acres, and the outlay about 10*l.* per acre. From 1719 to 1759, the average number of enclosure Acts passed was 8 a year; 1780 to 1794, it was 30; 1797 to 1803, it was 83; in 1811 it was 134, (the highest number known); in 1814, 119; in 1816, 49; in 1827, 21; in 1829, 24; and in 1831, only 10. The great extent to which the enclosure system thus appears to have already been carried, now necessarily diminishes the progress of enclosures every year.

(22.) Among the various causes of the superiority of English husbandry over that of the Continent, is that of the medium size of our farms, which differ both from such large unmanageable tracts as those held by Polish noblemen, and from such diminutive occupancies as those which have prevailed in France since the first Revolution, in consequence of the abolition of the law of primogeniture. The size of English farms is the greatest in the best cultivated districts; such as Kent, Essex, Suffolk, Norfolk, and Northumberland. In these counties the engagements of the farmers are very large, and frequently amount to 1000*l.* a year and upwards. In more retired districts, as in Cumberland, Westmorland, and Wales, the occupancies are, in general, small, and an average of all the farms in England and Wales would, perhaps, not exceed 150*l.* a year. Leases are, for the most part, granted for seven years only, and farms are occasionally let from year to year upon written agreements, with specified covenants subjecting the tenants to fines in the event of deviation from them. The tenants of great landholders, particularly of the old nobility, often hold at will, without leases, upon the understanding of conformity to the rules laid down by the lord for the observance of all his tenants; and such tenants are found to occupy from father to son for many generations. Upon the whole, the tenure of leasehold property in England is considered to be too short to admit of the improvements that tenants might otherwise be expected to make in our system of agriculture.

(23.) The expense of cultivation of land in England has much increased of late years, as appears by the returns to the Board of Agriculture, which state that the average expenses of cultivating one hundred acres of land was in 1790, 411*l.*; in 1803, 547*l.*; and in 1813, 771*l.* Since the latter year there have been reductions in labour and taxes, and also, to a considerable extent, in rent. Surveyors calculate that highly cultivated land ought to produce a

threefold return, viz.: one-third of the gross produce to the landlord for rent, another for the expenses, and the remainder for the farmer's profit; the rent of inferior land being only a fourth, or even a fifth of the gross produce, by reason of the additional expense of cultivation.

(24.) A century ago, our cattle, from the inferiority of their feed, were not one-half, sometimes even not one-third, of their present weight. It is computed that England and Wales now contain, at least, five million oxen, and a million and a half of horses, of which about a million are used in husbandry, 200,000 for pleasure, and 300,000 are colts and breeding mares. The number of sheep is about twenty millions, and eight million lambs. The number of long-wooled sheep is about five millions, their fleeces averaging 7 or 8 lbs.; and of short-wooled sheep fifteen millions, the weight of fleece averaging from 3 to 3½ lbs. The whole quantity of wool annually shorn in England is from eighty to eighty-five million of pounds. The Merino were introduced about the beginning of the present century, and were imported in large numbers after our alliance with Spain in 1809. The great pasturage counties are Leicester, Northampton, Lincoln, and Somerset; and for butter and cheese, Cheshire, Gloucestershire, and Wiltshire. The import of butter and cheese from foreign countries is checked by duties, but these are important articles of Irish commerce with England.

(25) The annual amount of profit from farming is not very susceptible of exact calculation, but was estimated some fifteen years since at thirty millions sterling, being a sum equivalent to the rental of England and Wales. The probable amount of the farming capital of the country was estimated at from two hundred and fifty to three hundred millions sterling. In regard to the value of the total annual produce of the land, this is necessarily subject to the fluctuations of seasons, but taking wheat at the medium of 80s. and other corn in proportion, we shall find an average produce of more than sixty millions sterling in corn, to which adding a similar value in pasturage, and a further allowance for hops, fruit, and vegetables, we have a total of from 130 to 140 millions. In Scotland the rent bears a higher proportion to the gross produce, being, in general, not less than one-third. Our chief superiority over the Continent consists in machinery and live stock. Much valuable information on the state of agriculture on the Continent is to be found in the Reports to the Government of Mr. Jacob, who travelled a few years since with a view of ascertaining the effect that would be produced by the modification of our

corn-laws. From these Reports it seems that the difficulties of transport in the corn countries, and other impediments to production, are such as to render the probable extent of importation under a more free system much less than is commonly imagined. There are many improvements of which English agriculture is susceptible, such as in the size of farms in many counties, the length of leases, the course of husbandry, the construction of ploughs, and the misapplication of animal strength in labour. With attention to these points and the application of further capital, not to wastes, but to fertile land already under culture, there is every hope that our agriculture may be yet considerably advanced in productiveness and in national value.

[To be continued.]

An officer in the forty-fourth regiment, who had occasion, when in Paris, to pass one of the bridges across the Seine, had his boots, which had been previously well-polished, dirtied by a poodle-dog rubbing against them. He, in consequence, went to a man who was stationed on the bridge, and had them cleaned. The same circumstance having occurred more than once, his curiosity was excited, and he watched the dog. He saw him roll himself in the mud of the river, and then watch for a person with well-polished boots, against which he contrived to rub himself. Finding that the shoe-black was the owner of the dog, he taxed him with the artifice; and after a little hesitation, he confessed that he had taught the dog the trick in order to procure customers for himself. The officer being much struck with the dog's sagacity, purchased him at a high price, and brought him to England. He kept him tied up in London some time, and then released him. The dog remained with him a day or two, and then made his escape. A fortnight afterwards he was found with his former master, pursuing his old trade on the bridge.—*Jesse's Gleanings of Natural History*

THE MAHOGANY TABLE.

Milton, who was at once the most sublime and the most practical of writers, has said,

—— “To know
That which before us lies in daily life,
Is the prime wisdom.”

The poet more especially had in view that knowledge to which all other knowledge is secondary—we mean the knowledge of ourselves. But we may not improperly adopt his forcible expressions as a motto to a series of articles which we shall occasionally publish, which will have for their object to collect some of the most striking facts belonging to the commonest things by which we are surrounded in our every-day life, particularly those comforts and conveniences which the humblest man possesses in a state of advanced civilization. The history of a knife, or a button, or a coat, or a watch, or an earthen pan, or a candle, or a lump of coal, or a mahogany table, or a Penny Magazine, suggests to our minds more precise and satisfactory notions of the progress of society, and therefore of the real history of the people of these kingdoms, than all the details of wars and treaties and state intrigues, of which history is in general made up. In the execution of such a purpose it is not important to pursue any systematic plan. The most material consideration will be to select those things of ordinary use which are so common, that it would be difficult to find a single reader who is not more or less indebted to them for some of his enjoyments.

We will begin with a Mahogany Table. If we had been speaking about a mahogany table, or any other article of mahogany, thirty or forty years ago, we should have expected only to have interested the rich in the description of this important material of English furniture. Now, what tradesman, or mechanic, or even cottager, does not possess some article of mahogany—if it be only a tea-caddy? The universal employment of mahogany for articles of furniture, whose price does not operate as a prohibition against their use in general society, has been produced by the large application of capital to the commercial speculation of bringing mahogany logs to this country from the West Indies,—and, further, by the invention of machinery for cutting those logs into thin layers, called veneers, by which operation the finest wood is brought within a reasonable cost. Now observe what commercial enterprise and mechanical ingenuity will accomplish in a comparatively

small period. Some piece of mahogany furniture is now, probably, found in every house in England;—a hundred and eight years ago the wood was unknown here. A physician of the name of Gibbons, who resided in London, received in 1724 a present of some mahogany planks from his brother, a West-India captain. Dr. Gibbons was then building a house in King-street, Covent-garden, and he desired his carpenter to work up the wood. The carpenter had no tool hard enough to touch it; so the planks were laid aside. The doctor's wife, after the house was finished, wanted a candle-box, and the mahogany was again thought of. A cabinet-maker of the name of Wollaston was applied to; and he also complained that his tools were too soft. But he persevered, and the candle-box was at length completed—after a rude fashion no doubt. The candle-box was so much admired, that the physician resolved to have a mahogany bureau; and when the bureau was finished, all the people of fashion came to see it. The cabinet-maker procured more planks, and made a fortune by the numerous customers he obtained. From that time the use of mahogany furniture went forward amongst the luxurious;—and the drawers and bureaus of walnut-tree and pear-tree were gradually superseded in the houses of the rich. To show the present extensive use of mahogany in this country it may be sufficient to mention that in 1829 the importation of this wood amounted to 19,335 tons.

The common mahogany (called by botanists *Swietenia mahagoni*) is one of the most majestic trees of the whole world. There are trees of greater height than the mahogany;—but in Cuba and Honduras this tree, during a growth of two centuries, expands to such a gigantic trunk, throws out such massive arms, and spreads the shade of its shining green leaves over such a vast surface, that even the proudest oaks of our forests appear insignificant in comparison with it. A single log, such as is brought to this country from Honduras, not unfrequently weighs six or seven tons.



[Mahogany Tree.]

When we consider the enormous size of a trunk of mahogany, and further learn that the most valuable timber grows in the most inaccessible situations, it must be evident that a great portion of the price of this timber must be made up of the cost of the labour required for transporting it from its native forests to the place of its embarkation for England. The mode in which this difficult work is accomplished is highly interesting; and we have, fortunately, the means of giving an account of the process (which, we believe, has never before been described in any English publication,) from some statements printed in a Honduras Almanac, which has been kindly put into our hands for this purpose.

The season for cutting the mahogany usually commences about the month of August. The gangs of labourers employed in this work consist of from twenty to fifty each, but few exceed the latter number. They are composed of slaves and free persons, without any comparative distinction of rank, and it very frequently occurs that the conductor of such work, here styled the Captain, is a slave. Each gang has also one person belonging to it termed the Huntsman. He is generally selected from the most intelligent of his fellows, and his chief occupation is to search the woods, or, as it is called, *the bush*, to find labour for the whole. Accordingly, about the beginning of August, the huntsman is despatched on his important mission. He cuts his way through the thickest of the woods to some elevated situation, and climbs the tallest tree he finds, from which he minutely surveys the surrounding country. At this season the leaves of the mahogany tree are invariably of a yellow reddish hue, and an eye accustomed to this kind of exercise, can, at a great distance, discern the places where the wood is most abundant. He now descends, and to such places his steps are directed; and, without compass, or other guide than what observation has imprinted on his recollection, he never fails to reach the exact point at which he aims. On some occasions no ordinary stratagem is necessary to be resorted to, by the huntsman, to prevent others from availing themselves of the advantage of his discoveries; for, if his steps be traced by those who may be engaged in the same pursuit, which is a very common thing, all his ingenuity must be exerted to beguile them from the true scent. In this, however, he is not always successful, being followed by those who are entirely aware of all the arts he may use, and whose eyes are so quick that

the lightest turn of a leaf, or the faintest impression of the foot, is unerringly perceived. The treasure being, however, reached by one party or another, the next operation is the felling of a sufficient number of trees to employ the gang during the season. The mahogany tree is commonly cut about ten or twelve feet from the ground, a stage being erected for the axe-man employed in levelling it. The trunk of the tree, from the dimensions of the wood it furnishes, is deemed the most valuable; but, for ornamental purposes, the limbs, or branches, are generally preferred.

A sufficient number of trees being felled to occupy the gang during the season, they commence cutting the roads upon which they are to be transported. This may fairly be estimated at two-thirds of the labour and expense of mahogany cutting. Each mahogany work forms in itself a small village on the bank of a river,—the choice of situation being always regulated by the proximity of such river to the mahogany intended as the object of future operations.

After completing the establishment of a sufficient number of huts for the accommodation of the workmen, a main road is opened from the settlement, in a direction as near as possible to the centre of the body of trees so felled, into which branch-roads are afterwards introduced, the ground through which the roads are to run being yet a mass of dense forest, both of high trees and underwood. The labourers commence by clearing away the underwood with cutlasses. This labour is usually performed by task-work, of one hundred yards, each man, per day. The underwood being removed, the larger trees are then cut down by the axe, as even with the ground as possible, the task being also at this work one hundred yards per day to each labourer. The hard woods growing here, on failure of the axe, are removed by the application of fire. The trunks of these trees, although many of them are valuable, such as bullet-tree, ironwood, redwood, and sapodilla, are thrown away as useless, unless they happen to be adjacent to some creek or small river, which may intersect the road. In that case they are applied to the construction of bridges, which are frequently of considerable size, and require great labour to make them of sufficient strength to bear such immense loads as are brought over them.

If the mahogany trees are much dispersed or scattered, the labour and extent of road-cutting is, of course, greatly increased. It not unfrequently occurs that miles of road and many bridges are made to a single tree, that

may ultimately yield but one log. When roads are cleared of brush-wood, they still require the labour of hoes, pick-axes, and sledge hammers to level down the hillocks, to break the rocks, and to cut such of the remaining stumps as might impede the wheels that are hereafter to pass over them.

The roads being now in a state of readiness, which may generally be effected by the month of December, the cross cutting, as it is technically called, commences. This is merely dividing crosswise, by means of saws, each mahogany tree into logs, according to their length; and it often occurs, that while some are but long enough for one log, others, on the contrary, will admit of four or five being cut from the same trunk or stem. The chief guide for dividing the trees into logs is the necessity for equalizing the loads the cattle have to draw. Consequently, as the tree increases in thickness, the logs are reduced in length. This, however, does not altogether obviate the irregularity of the loads, and a supply of oxen are constantly kept in readiness to add to the usual number, according to the weight of the log. This becomes unavoidable, from the very great difference of size of the mahogany trees, the logs taken from one tree being about 300 cubic feet, while those from the next may be as many thousand. The largest log ever cut in Honduras was of the following dimensions:—Length, 17 feet; breadth, 57 inches; depth, 64 inches; measuring 5,168 superficial feet, or 15 tons weight.

The sawing being now completed, the logs are reduced, by means of the axe, from the round or natural form, into the square. The month of March is now reached, when all the preparation before described is, or ought to be, completed; when the dry season, or time of drawing down the logs from the place of their growth, commences. This process can only be carried on in the months of April and May; the ground, during all the rest of the year, being too soft to admit of a heavily laden truck to pass over it without sinking. It is now necessary that not a moment should be lost in drawing out the wood to the river.

A gang of forty men is generally capable of working six trucks. Each truck requires seven pair of oxen and two drivers; sixteen to cut food for the cattle, and twelve to load or put the logs on the carriages. From the intense heat of the sun, the cattle, especially, would be unable to work during its influence; and, consequently, the loading and carriage of the timber is performed in the night. The logs are placed upon the trucks by means of a

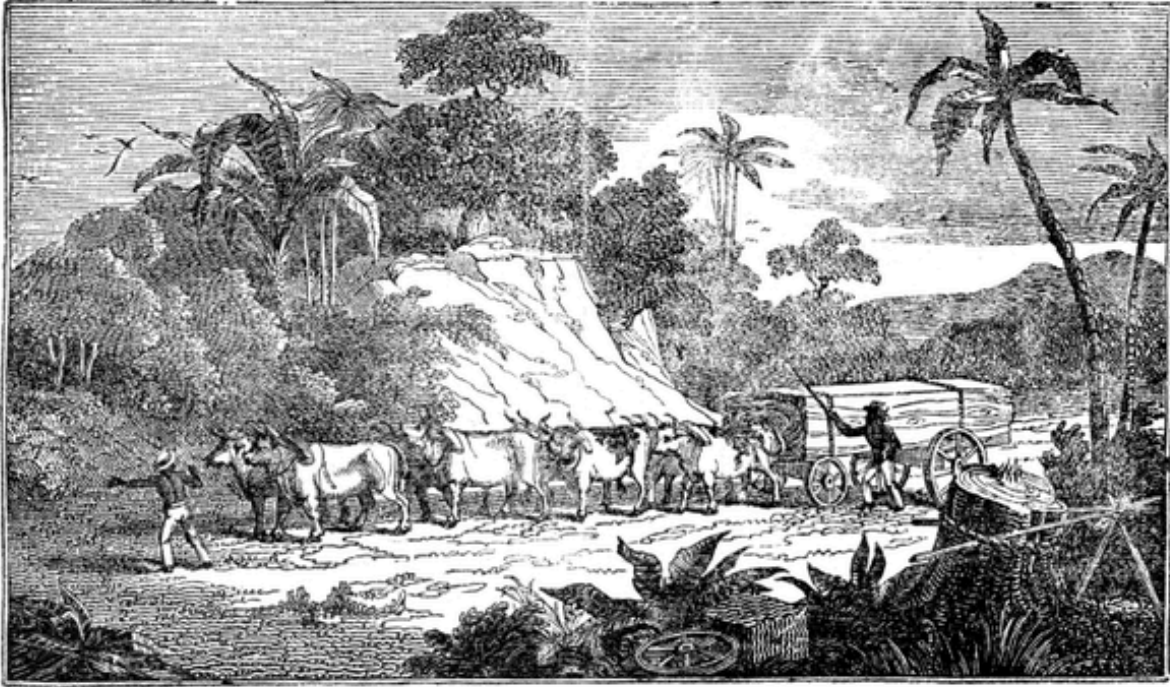
temporary platform laid from the edge of the truck to a sufficient distance upon the ground, so as to make an inclined plane, upon which the log is gradually pushed up by bodily labour, without any further mechanical aid.

The operations of loading and carrying are thus principally performed during the hours of darkness. The torches employed are pieces of wood split from the trunk of the pitch-pine. The river-side is generally reached by the wearied drivers and cattle before the sun is at its highest power; and the logs, marked with the owner's initials, are thrown into the river.

About the end of May the periodical rains again commence; the torrents of water discharged from the clouds are so great as to render the roads impassable in the course of a few hours, when all trucking ceases. About the middle of June the rivers are swollen to an immense height. The logs then float down a distance of two hundred miles, being followed by the gang in pitpans (a kind of flat-bottomed canoe), to disengage them from the branches of the overhanging trees, until they are stopped by a boom placed in some situation convenient to the mouth of the river. Each gang then separates its own cutting, by the marks on the ends of the logs, and forms them into large rafts; in which state they are brought down to the wharves of the proprietors, where they are taken out of the water, and undergo a second process of the axe, to make the surface smooth. The ends, which frequently get split and rent by being dashed against rocks in the river by the force of the current, are also sawed off. They are now ready for shipping.

The ships clearing out from Belize, the principal port of Honduras, with their valuable freight of mahogany, either come direct to England, or take their cargo to some free warehousing port of the British possessions in the West Indies or America.

We must describe the beautiful process of cutting mahogany logs into veneers, before we have reached the point when the skill of the cabinet-maker is employed to produce a mahogany table. This shall be done in an early number.



[Trucking Mahogany.]

WHAT IS EDUCATION?

This may seem a very simple question, and very easily answered; but many who think so, would really be very much at a loss to answer it correctly. Every man, in a free country, wants three sorts of education:—one, to fit him for his own particular trade or calling,—this is professional education;—another, to teach him his duties as a man and a citizen,—this is moral and political education;—and a third, to fit him for his higher relations, as God’s creature, designed for immortality,—this is religious education. Now, in point of fact, that is most useful to a man which tends most to his happiness; a thing so plain, that it seems foolish to state it. Yet people constantly take the word “useful” in another sense, and mean by it, not what tends most to a man’s happiness, but what tends most to get money for him; and therefore they call professional education a very useful

thing: but the time which is spent in general education, whether moral or religious, they are apt to grudge as thrown away, especially if it interferes with the other education, to which they confine the name of “useful;” that is, the education which enables a man to gain his livelihood. Yet we might all be excellent in our several trades and professions, and still be very ignorant, very miserable, and very wicked. We might do pretty well just while we were at work on our business; but no man is at work always. There is a time which we spend with our families; a time which we spend with our friends and neighbours; and a very important time which we spend with ourselves. If we know not how to pass these times well, we are very contemptible and worthless *men*, though we may be very excellent lawyers, surgeons, chemists, engineers, mechanics, labourers, or whatever else may be our particular employment. Now, what enables us to pass these times well, and our times of business also, is not our *professional* education, but our *general* one. It is the education which all need equally—namely, that which teaches a man, in the first place, his duty to God and his neighbour; which trains him to good principles and good temper; to think of others, and not only of himself. It is that education which teaches him, in the next place, his duties as a citizen—to obey the laws always, but to try to get them made as perfect as possible; to understand that a good and just government cannot consult the interests of one particular class of calling, in preference to another, but must see what is for the good of the whole; that every interest, and every order of men, must give and take; and that if each were to insist upon having everything its own way, there would be nothing but the wildest confusion, or the merest tyranny. And because a great part of all that goes wrong in public or private life arises from ignorance and bad reasoning, all that teaches us, in the third place, to reason justly, and puts us on our guard against the common tricks of unfair writers and talkers, or the confusions of such as are puzzle-headed, is a most valuable part of a man’s education, and one of which he will find the benefit whenever he has occasion to open his mouth to speak, or his ears to hear. And, finally, all that makes a man’s mind more active, and the ideas which enter it nobler and more beautiful, is a great addition to his happiness whenever he is alone, and to the pleasure which others derive from his company when he is in society. Therefore it is most *useful* to learn to love and understand what is *beautiful*, whether in the works of God, or in those of man; whether in the flowers and fields, and rocks and woods, and rivers, and sea and sky; or in

fine buildings, or fine pictures, or fine music; and in the noble thoughts and glorious images of poetry. This is the education which will make a man and a people good, and wise, and happy. Give this,—and the ends of professional education can never be altogether lost; for good sense and good principle will ensure a man's knowing his particular business; but knowledge of his business, on the other hand, will not ensure *them*; and not only are sense and goodness the rarest and most profitable qualities with which any man can enter upon life now, but they are articles of which there never can be a glut: no competition or over-production will lessen their value; but the more of them that we can succeed in manufacturing, so much the higher will be their price, because there will be more to understand and to love them.

Honesty is the best Policy.—Irritated one day at the bad faith of Madame Jay, Mirabeau said to her in my presence, “Madam Jay, if probity did not exist, we ought to invent it, as the best means of getting rich.”—*Dumont*

THE WEEK.



[The Rev. John Wesley.]

June 17.—The birth-day of JOHN WESLEY, the celebrated founder of the more numerous division of the English Methodists. He was the second son of the Rev. Samuel Wesley, Rector of Epworth, in Lincolnshire, where he was born in the year 1703. Although his father was a man of considerable literary attainments, being known to the public as the author of various

works in verse, it was to his mother, a woman of a much more zealous and active character than her husband, that Wesley was chiefly indebted for his early education, and probably also for the seeds of many of his distinguished mental habits.

After receiving a very systematic elementary tuition from his mother, John Wesley was sent to the Charterhouse, from whence he removed at the usual time to Christ-church College, Oxford. Here he distinguished himself greatly by his diligence and success as a student, showing from the first, in the distribution of his time, the same punctual and persevering regard to method by means of which he mainly achieved all the greater objects of his life. The reading of some religious works, and especially of 'Law's Serious Call,' awakened in him a strong spirit of religious fervour; and he formed that association with a number of his college acquaintances of similar views and feelings, to which, from the punctilious regularity of the members in their devotions and general demeanour, the epithet of "methodists" was given as a name of reproach by the wags of the university. As has happened in other cases, the objects of the intended satire were much too earnest in the views they had adopted to feel or to regard any point of ridicule which it might be supposed to possess, and frankly adopted the nick-name thus bestowed upon them by their opponents, as their proper designation. Among their number, besides Wesley, was the afterwards equally celebrated George Whitfield.

We cannot here attempt to pursue minutely the remainder of the course of Wesley's busy life, or to trace the rise of that extensive fabric of ecclesiastical policy of which he was the founder. Suffice it to say, that having commenced his public labours as a religious teacher in the newly-formed colony of Georgia, in America, in the year 1735, he pursued from this time a course of almost constant journeying, preaching, and writing, till within a week of his death, on the 2d of March, 1791, in the eighty-eighth year of his age. During the greater part of this long period he rarely preached less than twice, and often four or five times a day; while, besides presiding with the most minute superintendence over all the public affairs of the large and rapidly growing community which acknowledged him as its head, and transacting a great deal of private business, he found time to send to the press a succession of works, which, in the collected edition, amount to between thirty and forty volumes. Mr. Southey, who has made the life of this extraordinary man one of the most interesting books in the language,

has given us the following account of the manner in which he contrived to get through all this occupation. "Leisure and I," said Wesley, "have taken leave of one another. I propose to be busy as long as I live, if my health is so long indulged to me." This resolution was made in the prime of life, and never was resolution more punctually observed. "Lord, let me not live to be useless!" was the prayer which he uttered after seeing one whom he had long known as an active and useful magistrate, reduced by age to be "a picture of human nature in disgrace, feeble in body and mind, slow of speech and understanding." He was favoured with a constitution vigorous beyond that of ordinary men, and with an activity of spirit which is even rarer than his singular felicity of health and strength. Ten thousand cares of various kinds, he said, were no more weight or burthen to his mind than ten thousand hairs were to his head.... His manner of life was the most favourable that could have been devised for longevity. He rose early, and lay down at night with nothing to keep him waking, or trouble him in sleep. His mind was always in a pleasurable and wholesome state of activity; he was temperate in his diet, and lived in perpetual locomotion. And frequent change of air is, perhaps, of all things, that which most conduces to joyous health and long life. The time which Mr. Wesley spent in travelling was not lost. "History, poetry, and philosophy," said he, "I commonly read on horseback, having other employment at other times." He used to throw the reins on his horse's neck, and in this way he rode, in the course of his life, above a hundred thousand miles, without any accident of sufficient magnitude to make him sensible of the danger which he incurred.

June 21.—*The Longest Day*.—On this day there is an interval of sixteen hours and thirty-four minutes between the rising and the setting of the sun, which interval is longer than on any other day in the year. Up to this point, from the 21st December (the shortest day), the days have steadily increased in length; from this point they will steadily decrease. We may more properly, at some future time, explain in a series of papers some of the more remarkable phenomena of the changes of seasons. At present we shall call our reader's attention to the moral reflections which the recurrence of "The Longest Day" suggests, by re-printing a few stanzas of a poem by Mr. Wordsworth on this subject:—

Summer ebbs;—each day that follows
Is a reflux from on high,
Tending to the darksome hollows

Where the frosts of winter lie.

He who governs the creation,
In his providence assign'd
Such a gradual declination
To the life of human kind.

Yet we mark it not;—fruits redden,
Fresh flowers blow, as flowers have blown,
And the heart is loth to deaden
Hopes that she so long hath known.

Be thou wiser, youthful Maiden!
And when thy decline shall come,
Let not flowers, or boughs fruit-laden,
Hide the knowledge of thy doom.

Now, even now, ere wrapp'd in slumber.
Fix thine eyes upon the sea
That absorbs time, space, and number;
Look towards eternity!

Follow thou the flowing river,
On whose breast are thither borne
All deceived, and each deceiver,
Through the gates of night and morn;

Through the year's successive portals;
Through the bounds which many a star
Marks, not mindless of frail mortals,
When his light returns from far.

Thus when Thou with Time hast travell'd
Tow'rds the mighty gulf of things,
And the mazy stream unravell'd
With thy best imaginings;

Think, if thou on beauty leanest,

Think how pitiful that stay,
Did not virtue give the meanest
Charms superior to decay.

Duty, like a strict preceptor,
Sometimes frowns, or seems to frown;
Choose her thistle for thy sceptre,
While thy brow youth's roses crown.

GENIUS AND INDUSTRY.

Whilst we believe that education is the greatest gift that can be conferred on a human creature, we are not sanguine enough to expect that its more general diffusion will increase the number of men of genius. There is a perversity in human nature which makes us relax our efforts at the moment when they might be rewarded with the most splendid success. It does not follow that a shepherd-boy, who passes his long day on the side of a hill, and who acquires the principles of mechanics, or forms for himself a plan of the stars, shall make proportionate advancement if full opportunity of study be afforded to him.

Nor does it follow that a young man who teaches himself to read by the light of a shop window in the street, shall become a learned man when admitted to libraries and encouraged by applause.

We do not think the illustration a correct one, which represents the scholar as like the weary traveller who plods on contentedly through woods and over irregular ground which conceal the prospect, and who faints when he has ascended to the top of the hill and sees the whole extent of the road before him.

The truth seems rather to be, that energy of mind, like strength of body, must be acquired by exercise, and that the consciousness of desert in encountering difficulties, must be felt to enable us to accomplish any great work. Sir Joshua Reynolds has happily expressed this:—

“It is not uncommon to see young artists, whilst they are struggling with every obstacle in their way, exert themselves with such success as to outstrip competitors possessed of every means of improvement. The promising expectation which was formed on so much being done with so little means, has recommended them to a patron, who has supplied them with every convenience of study; from that time their industry and eagerness of pursuit have forsaken them; they stand still and see others rush on before them.

“Such men are like certain animals, who will feed only where there is little provender, and that got at with difficulty through the bars of a rack, but refuse to touch it where there is an abundance before them^[2].”

From this it appears to be essential to success that a young man should study to acquire confidence in his own powers. This is a condition of mind entirely different from conceit; it exhibits itself in no vain boasting, but essentially consists in a secret resolution to make great efforts by persevering industry, to gain the object of his ambition.

We believe that young men would entertain these notions oftener, if they were not deterred by an erroneous fancy of what belongs to genius. They think that such exertions as we recommend belong only to a plodding fellow, whilst the man of genius does every thing by a sudden act which costs him nothing.

This is an unhappy mistake. All our eminent men have been distinguished by fixing upon some great object, and possessing themselves with such a lively conception of it that it has led them on through years of toil.

². Sir J. Reynolds’ Works, vol. ii. p. 80.

Every one says that geography is one of the most useful things that can be learnt; yet nothing is learnt so ill, because nothing is taught so ill. Look into any of the elementary books of geography, and read what is said about England. First, we are told that it is divided into forty counties; then, perhaps, follows an account of the several law circuits; and then, after some short notices about religion, government, produce, and manufactures, there are given lists of the chief towns, mountains, rivers, and lakes. But all these things are given without any connexion with each other, and it is a mere matter of memory to recollect what is no more than a string of names. And if a man does recollect them, still he is not much the wiser for them; he has got no clear and instructive notions about the country, but has merely learnt his map, and knows where to find certain names and lines upon it.

If we wish to know geography really, we must set about it in a very different manner. Take one of the skeleton maps published by the Useful Knowledge Society; there is not a single name upon them, nothing is given but the hills and the rivers. These are the true alphabet of geography. The hills are the bones of a country, and determine its form, just as the bones of an animal do. For according to the direction of the hills must be the course of the rivers: if the hills come very near the sea, it makes the rivers very short and their course very rapid; if they are a long way from the sea, it makes the rivers long and gentle. But rivers of this latter sort are generally navigable, and become so large near the sea as to be capable of receiving ships of large size. Here then towns will be built, and these towns will become rich and populous, and so will acquire political importance. Again, on the nature of the hills depend the mineral riches of a country; if they are composed of granite or slate, they may contain gold, silver, tin, and copper; if they are composed of the limestone of Derbyshire or Durham, they are very likely to have lead mines; if of the sand or gritstone of Northumberland, Lancashire, and Yorkshire, it is probable that there will be coal at no great distance. On the contrary, if they are made up of the yellow limestone of Gloucestershire, Oxfordshire, and Northamptonshire, or of chalk like the hills in Wiltshire, Berkshire, and Hampshire, or of clay like those about London, it is quite certain that they will contain neither coal, nor lead, nor any valuable mineral whatsoever. But on the mineral wealth of a country, and particularly on its having coal or not having it, depends the nature of the employment of its inhabitants. Manufactories are sure to follow coal mines; whereas, in all those districts of England where there is

no coal, that is, in all the counties to the south-east of a line drawn from the Wash in Lincolnshire to Plymouth, there are, generally speaking, no manufactories; but the great bulk of the people are employed in agriculture.

Thus then on the direction and composition of the hills of a country depend, first of all, the size and character of its rivers. On the character of its rivers depend the situation and importance of its towns, and its greater or less facilities for internal communication and foreign trade. And again, on the composition of the hills depend the employment of the people, their numbers on a given space, and in a great degree their state of morals, intelligence, and political independence. And here we have a reason for things, and see them connected with one another in a manner at once easier to remember, and much more satisfactory to understand when we do remember it. Some instances of this, given in detail, may appear in one of our future numbers.

The Flower Garden (June).—It will now be time for you to take up those bulbs, of which the leaves are nearly decayed. I can fix no particular day for this operation; because, as the bulbs flower at different seasons, so the leaves will decay in like manner; but the general rule is, to take them up carefully as soon as the leaves have turned yellow, and to lay them under a south wall to dry and ripen; taking care to cover them with fine, dry, sandy earth, in layers, so that they may not touch each other. When the leaves are quite decayed, the bulbs must be removed, and spread again to dry under shelter of a green-house, or in a room; and, finally, after cleaning them from the dirt, take off their old coats, or skins, and put them away in bags, or drawers, in a cool dry place, till they are wanted for replanting in the autumn. I must here explain why bulbs are taken up every year: the great object is in this, as in all other operations of gardening, to imitate Nature; to make the existence of foreign plants as near as it can be to what they enjoy in their native place. Tulips, hyacinths, and most of those bulbs which are taken up, come from countries where the whole summer is dry, and in winter the ground is covered with snow; the spring rains alone call them into life and flower. Travellers describe whole regions in Persia as being covered in the spring with enamelled carpets of scilla (hyacinths), tulips,

and other bulbous plants: long drought succeeds the rains of spring, the leaves die away, and the plant rests again under the dry earth till the following spring. As in our country they can have no dry earth naturally to rest in during the summer, the best imitation of it is to take up the bulb, which would otherwise be rotted by the summer rains, or caused to grow in the autumn; in which latter case, the plant would not flower in the spring, as the flower-stalks would be killed by the wet and cold of winter, before it came to the surface.

* * From 'The Garden,' a very agreeable and instructive book for children, forming one of the volumes of a series called 'The Little Library.'

"A little Learning is a dangerous Thing."—Then make it greater. No learning at all is surely the most dangerous thing in the world; and it is fortunate that, in this country at least, it is a danger which cannot possibly exist. After all, learning is acquired knowledge, and nothing else. A man who can read his Bible has a little learning; a man who can only plough or dig, has less; a man who can only break stones on the road, less still, but he has some. The savages in one of the islands in the South Sea, stood with great reverence round a sailor who had lighted a fire to boil some water in a saucepan, but as soon as the water began to boil, they ran away in an agony of terror. Compared with the savages, there is no boy in Europe, of the age of ten years, who may not be called learned. He has acquired a certain quantity of practical knowledge in physics; and, as this knowledge is more than instinct, it is learning; learning which differs in degree only from that which enables a chemist to separate the simple metals from soda or potash.

The geographer Malte Brun remarks, that in many cities of the United States, that which is called a mob scarcely exists. Now it will be found that in these Cities education has been unstintedly bestowed upon all classes, down to the very lowest.

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Transcriber's Notes

New original cover art included with this eBook is granted to the public domain. Itemized changes from the original text:

- [p. 106](#): Added faint or unprinted dash to “one-third” in phrase “were not one-half, sometimes even not one-third, of their present weight.”
- [p. 107](#): Replaced “o” with “of” in phrase “a series of articles which we shall occasionally publish.”
- [p. 109](#): Supplied missing “l” in “principal” in phrase “the principal port of Honduras.”
- [p. 109](#): Replaced “Charter house” with “Charterhouse” in phrase “John Wesley was sent to the Charterhouse.”

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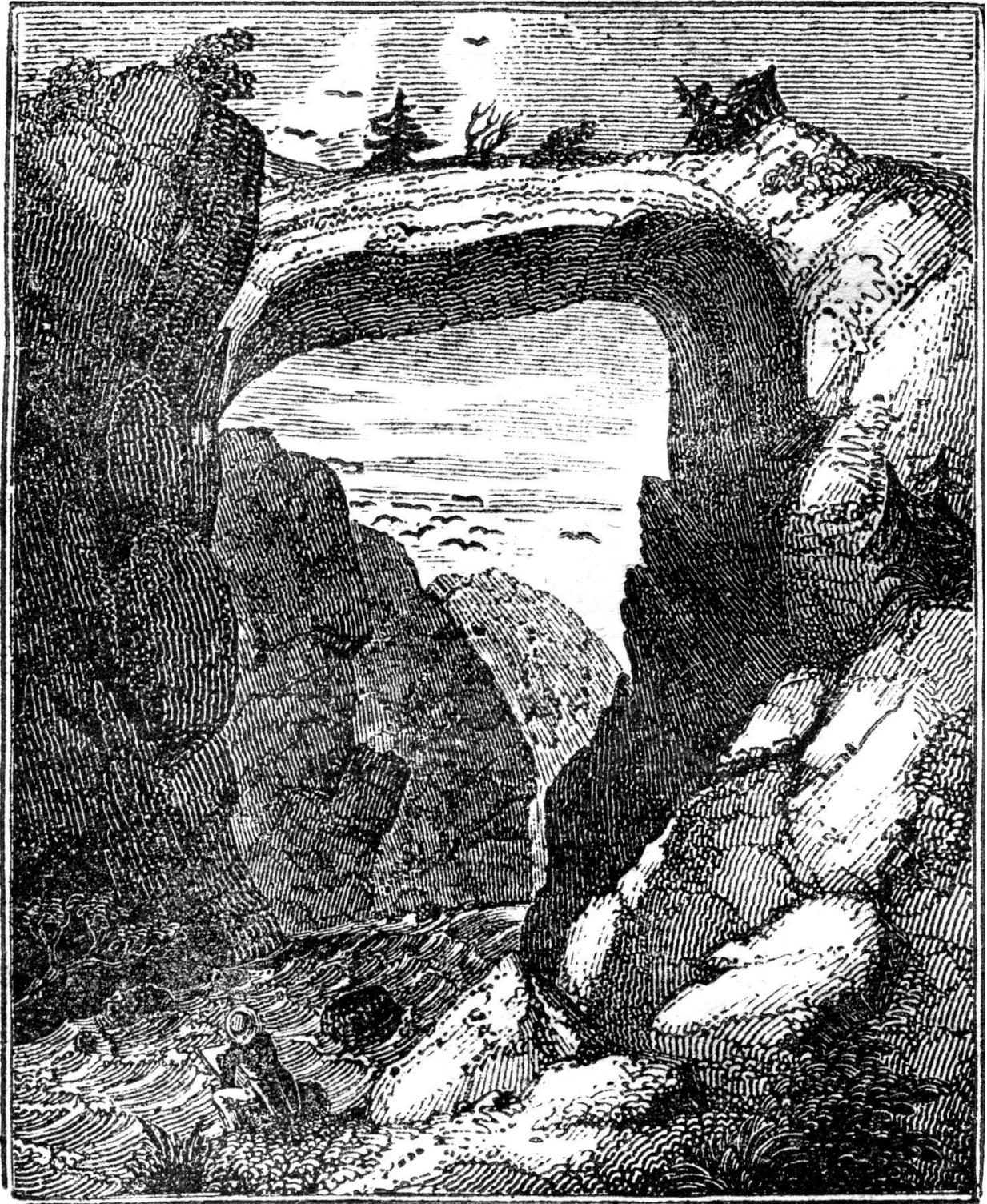
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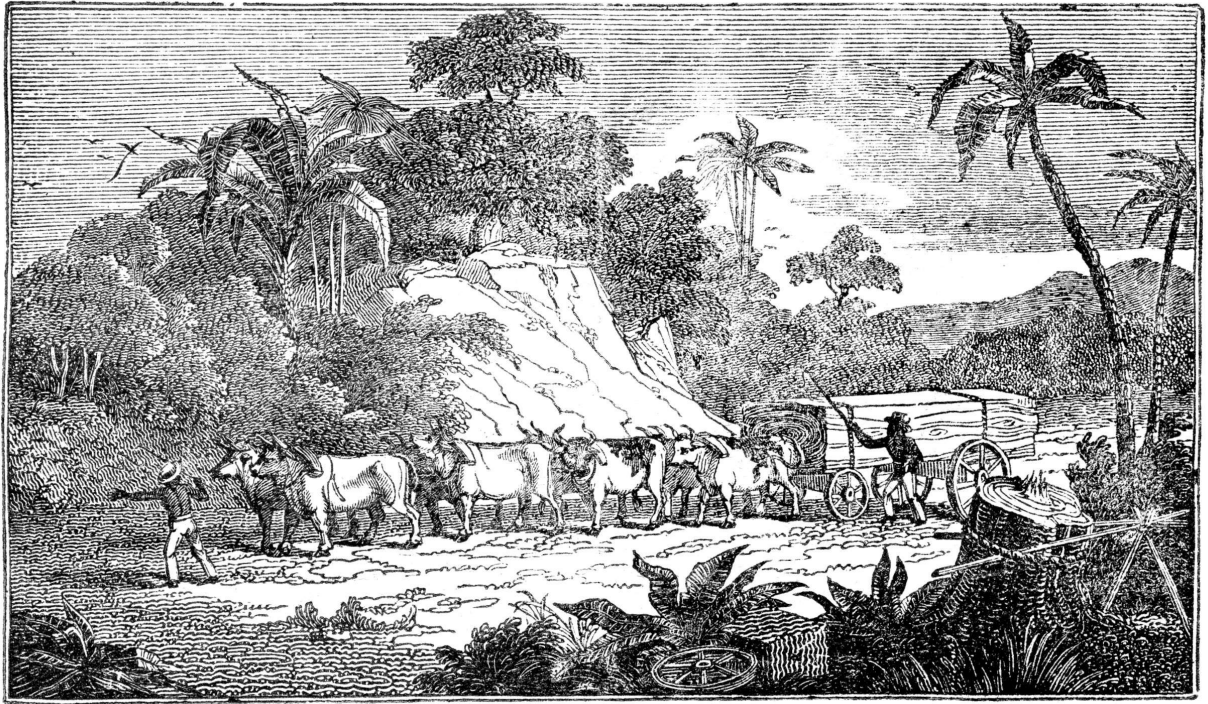
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